

Vectors

Student learning objectives for this unit

SUPPORTING SKILLS LEGEND

I — Introduce

D — Deepen

A — Apply

R — Reinforce

☐ 1. I can **represent** a vector geometrically and in component form.

RELATED SKILLS

☐ I Write a vector in component form.☐ A Add congruence, parallel, perpendicular, and angle markings to a diagram.

☐ 2. I can **add, subtract, and multiply** vectors by scalars.

RELATED SKILLS

☐ I Perform component-wise vector operations.☐ A Add, subtract, multiply, and divide signed integers.

☐ 3. I can **determine** vector magnitude, direction, and unit vector.

RELATED SKILLS

☐ I Calculate vector magnitude using the distance formula.☐ I Determine a vector's direction angle with correct quadrant.☐ I Normalize a nonzero vector to obtain a unit vector.☐ A Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula.

☐ 4. I can **resolve** a vector into horizontal and vertical components.

RELATED SKILLS

☐ D Write a vector in component form.☐ A Determine a vector's direction angle with correct quadrant.☐ A Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.

☐ 5. I can **perform** basic operations with vectors in three dimensions.

RELATED SKILLS

☐ I Represent and operate on vectors with three components.☐ D Calculate vector magnitude using the distance formula.☐ D Perform component-wise vector operations.